Chapter 1
Proofs by calculation

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you will learn

- Turn hypotheses into a chain that begins and ends at the required expressions.
- Separate algebraic identities from substitutions of known facts.
- Control the direction of inequalities and justify sign conditions.
- Translate individual steps into `calc`, `ring`, `rw`, `rel`, `numbers`, `extra`, and `addarith`.

A proof is an explanation

Every link must be justified, and the links must combine to establish the stated goal.

Start by identifying the number system

Domain	Examples	Lean name
$\mathbb{N}$	$0, 1, 2, \ldots$	`Nat`
$\mathbb{Z}$	$\ldots, -2, -1, 0, 1, \ldots$	`Int`
$\mathbb{Q}$	$-\frac{3}{2}, \frac{5}{7}, 4$	`Rat`
$\mathbb{R}$	$\frac{5}{7}, \sqrt{2}, \pi$	`Real`

- Here 0 is a natural number.
- The domain constrains the legal operations: $3 - 5$ in $\mathbb{N}$ is truncated subtraction.
- In calculations over $\mathbb{Q}$ or $\mathbb{R}$, division by a known nonzero constant is available.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 1 introduction and Section 1.3.

1.1 A chain proves its endpoints equal

Suppose each of the following statements has been justified:

$$E_0 = E_1, \qquad E_1 = E_2, \qquad E_2 = E_3.$$

Transitivity of equality gives $E_0 = E_3$.

A useful writing discipline

Start with the left side of the goal. End with its right side. Keep an equality sign between consecutive expressions.

Two common justifications:

- **Algebra:** the expressions agree for all values in the domain.
- **Substitution:** a hypothesis says that one expression equals another.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.1.

Worked example: reveal the expressions you know

Let $a, b \in \mathbb{Q}$, with $a - b = 3$ and $ab = 4$. Prove $(a + b)^2 = 25$.

$$\begin{aligned} (a+b)^2 &= (a-b)^2 + 4ab && \text{algebra} \\ &= 3^2 + 4 \cdot 4 && \text{the two hypotheses} \\ &= 25 && \text{arithmetic.} \end{aligned}$$

Why does the first step hold?

$(a - b)^2 + 4ab = a^2 - 2ab + b^2 + 4ab = a^2 + 2ab + b^2$.

The initial rewrite was chosen because $a - b$ and ab have known values.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.1; numerical variant of Example 1.1.1.

A calculation can avoid an unnecessary division

Let $a, b, c, d, e, f \in \mathbb{Z}$, with $ad = bc$ and $cf = de$. Then

$$\begin{aligned} d(af - be) &= (ad)f - dbe \\ &= (bc)f - dbe \\ &= b(cf) - dbe \\ &= b(de) - dbe = 0. \end{aligned}$$

Every step uses multiplication, subtraction, or a given equality.

Check the hypotheses before cancelling

Writing $a = bc/d$ would require $d \neq 0$ and a suitable domain. The hypotheses do not give $d \neq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 1.1.4.

Practice 1.1: write the chain before simplifying

1. Let $r, s \in \mathbb{R}$, with $r + 3s = 2$ and $s = -1$. Prove $r = 5$.
2. Let $a, b \in \mathbb{Q}$, with $a - b = 5$ and $ab = 6$. Prove $(a + b)^2 = 49$.

For each proof:

- State the first and last expressions before filling the middle.
- Mark every step as algebra, substitution, or arithmetic.

Practice 1.1: solution discussion

1. Introduce the expression $r + 3s$:

$$\begin{aligned} r &= (r + 3s) - 3s && \text{algebra} \\ &= 2 - 3(-1) && \text{substitution} \\ &= 5 && \text{arithmetic.} \end{aligned}$$

2. Use the same identity as in the worked example:

$$(a + b)^2 = (a - b)^2 + 4ab = 5^2 + 4 \cdot 6 = 49.$$

The calculation proves the requested expression directly; values of a and b are unnecessary.

1.2 Read a Lean statement as a mathematical sentence

```
example {a b : Rat}
    (hd : a - b = 3) (hp : a * b = 4) :
    (a + b) ^ 2 = 25 := by
```

- `a b : Rat`: the variables are rational numbers.
- `hd` and `hp`: names for the hypotheses.
- After the final colon: the proposition to prove.
- `by`: the proof will be given by tactics.
- Multiplication uses `*`; exponentiation uses `^`.

The examples use the textbook environment: its chapter file imports `Library.Basic` and runs `math2001_init`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.2 and accompanying Lean file.

The calculation, with one justification per line

```
example {a b : Rat}
    (hd : a - b = 3) (hp : a * b = 4) :
    (a + b) ^ 2 = 25 := by
  calc
    (a + b) ^ 2 = (a - b) ^ 2 + 4 * (a * b) := by ring
    _ = 3 ^ 2 + 4 * 4 := by rw [hd, hp]
    _ = 25 := by numbers
```

- `_` repeats the preceding right side.
- `ring` proves a polynomial identity.
- `rw [hd, hp]` substitutes the named equalities.
- `numbers` checks the final numerical equality.

Source: Heather Macbeth, *The Mechanics of Proof*, Method of Example 1.2.1.

Treat Lean feedback as a local question

At a line of a `calc` proof, ask:

1. What equality must this line establish?
2. Does it hold by algebra alone, or does it use a hypothesis?
3. If it uses a hypothesis, does its left side occur in the expression?

Parentheses matter to rewriting

$(ab)c$ and $a(bc)$ are equal, but have different syntactic structure. Use an algebraic step to expose the exact subexpression you plan to rewrite.

An unfinished placeholder is still an unfinished proof. Check that all proof obligations are resolved.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 1.2.2–1.2.4.

Practice 1.2: select the justification

Let $r, s \in \mathbb{R}$, with named hypotheses

$$\texttt{h1} : r + 3s = 2, \qquad \texttt{h2} : s = -1.$$

Fill the right column with `ring`, `rw [...]`, or `numbers`.

Step	Justification
$r = (r + 3s) - 3s$	_____
$(r + 3s) - 3s = 2 - 3s$	_____
$2 - 3s = 2 - 3(-1)$	_____
$2 - 3(-1) = 5$	_____

Which step would fail if you tried `ring` without using any hypothesis?

Practice 1.2: a complete Lean solution

```
example {r s : Real}
    (h1 : r + 3 * s = 2) (h2 : s = -1) : r = 5 := by
  calc
    r = (r + 3 * s) - 3 * s := by ring
    _ = 2 - 3 * s := by rw [h1]
    _ = 2 - 3 * (-1) := by rw [h2]
    _ = 5 := by numbers
```

`ring` cannot establish the substitution steps for arbitrary r, s. The hypotheses supply their mathematical content.

1.3 How to find the intermediate expressions

1. Write the target endpoints.
2. List useful expressions appearing in the hypotheses.
3. Rewrite the target to contain those expressions.
4. Substitute only after the matching expressions are visible.

A small example

If $x + 6 = 2$ in $\mathbb{Z}$, then

$$x = (x + 6) - 6 = 2 - 6 = -4.$$

Adding and subtracting 6 creates the expression we know.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; method of Example 1.3.2.

Worked example: combine two equations

Let $x, y \in \mathbb{Q}$, with $x + 2y = 9$ and $3x - y = 13$. Prove $x = 5$.

Planning

The coefficients of y are 2 and -1. Adding the first expression to twice the second cancels y.

$$\begin{aligned} x &= \frac{(x + 2y) + 2(3x - y)}{7} && \text{algebra} \\ &= \frac{9 + 2 \cdot 13}{7} && \text{substitution} \\ &= 5 && \text{arithmetic.} \end{aligned}$$

The denominator 7 is nonzero in $\mathbb{Q}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; methods of Examples 1.3.6–1.3.8.

Worked example: organize a polynomial around a hypothesis

Let $w, z \in \mathbb{R}$ and suppose $w + 2 = z$. Prove

$$w^2 + 5w + 4 = z^2 + z - 2.$$

$$\begin{aligned} w^2 + 5w + 4 &= (w+2)^2 + (w+2) - 2 && \text{algebra} \\ &= z^2 + z - 2 && \text{substitution.} \end{aligned}$$

Check the rewrite:

$$(w+2)^2 + (w+2) - 2 = w^2 + 4w + 4 + w + 2 - 2 = w^2 + 5w + 4.$$

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; method of Example 1.3.9.

Practice 1.3: create the expressions you need

1. If $x + y = 7$ and $2x - y = 2$ in $\mathbb{Q}$, prove $x = 3$.
2. If $a - 2 = b$ in $\mathbb{R}$, prove $a^2 - a + 1 = b^2 + 3b + 3$.

For the first problem, find a linear combination that cancels y.
For the second, rewrite the left side using $a - 2$.

Practice 1.3: solution discussion

1. The y coefficients already cancel:

$$x = \frac{(x+y)+(2x-y)}{3} = \frac{7+2}{3} = 3.$$

2. Complete a polynomial in $a-2$:

$$\begin{aligned} a^2 - a + 1 &= (a-2)^2 + 3(a-2) + 3 \\ &= b^2 + 3b + 3. \end{aligned}$$

Both proofs follow the same pattern: algebra first, substitution second.

1.4 An inequality needs a direction and a sign condition

For real numbers:

Known fact	Additional condition	Valid conclusion
$a \leq b$	none	$a + c \leq b + c$
$a \leq b$	none	$-b \leq -a$
$a \leq b$	$c \geq 0$	$ca \leq cb$
$a < b$	$c > 0$	$ca < cb$
$a \leq b$	$0 \leq a$	$a^2 \leq b^2$

Transitivity

$A \leq B < C \leq D$ implies $A < D$. A consistent chain with a strict step has a strict conclusion.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4.

Worked example: subtraction reverses the comparison

Let $x, y \in \mathbb{Z}$, with $x + 4 \leq 2$ and $y + 3x \geq 1$. Prove $y > 6$.

$$
\begin{aligned}
y &= (y + 3x) - 3x && \text{algebra} \\
&\geq 1 - 3x && y + 3x \geq 1 \\
&= 13 - 3(x + 4) && \text{algebra} \\
&\geq 13 - 3 \cdot 2 && x + 4 \leq 2 \\
&= 7 > 6 && \text{arithmetic.}
\end{aligned}
$$

When $x + 4$ becomes larger, $13 - 3(x + 4)$ becomes smaller. That is why the fourth line points toward $\geq$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.1.

Worked example: add a nonnegative square

Let $x, y \in \mathbb{R}$ and suppose $x^2 + y^2 \leq 2$. Prove $(x+y)^2 < 5$.

$$
\begin{aligned}
(x+y)^2 &\leq (x+y)^2 + (x-y)^2 && (x-y)^2 \geq 0 \\
&= 2(x^2+y^2) && \text{algebra} \\
&\leq 4 && \text{the hypothesis} \\
&< 5 && \text{arithmetic.}
\end{aligned}
$$

How was the extra square chosen?

Its expansion cancels the $2xy$ term and produces the known expression $x^2 + y^2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.8.

Valid individual comparisons can form an invalid proof

If $n \in \mathbb{Z}$ and $n \geq 6$, all these statements are true:

$$n^2 \geq 36 > 28 \leq 3n + 10.$$

They do **not** combine to prove $n^2 > 3n + 10$: the direction changes. For comparison, $4 > 3 \leq 100$ does not imply $4 > 100$.

A correct chain

Since $n \geq 6 > 0$,

$$n^2 = n \cdot n \geq 6n = 3n + 3n \geq 3n + 18 > 3n + 10.$$

Keep some dependence on n so that the final expression remains comparable.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.6.

Lean follows the same inequality reasoning

```
example {x y : Real} (h : x ^ 2 + y ^ 2 <= 2) :
    (x + y) ^ 2 < 5 := by
  calc
    (x + y) ^ 2 <= (x + y) ^ 2 + (x - y) ^ 2 := by extra
    _ = 2 * (x ^ 2 + y ^ 2) := by ring
    _ <= 2 * 2 := by rel [h]
    _ < 5 := by numbers
```

`extra`: elementary order facts such as adding a square.

`rel [h]`: transport a known comparison through compatible operations.

`rw` is for equality substitution; a hypothesis with $\leq$ needs order reasoning.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 1.4.7–1.4.8.

Practice 1.4: retain useful terms

1. Let $x \in \mathbb{R}$, with $x \geq 4$. Prove $x^2 - 2x \geq 8$.
2. Let $a, b \in \mathbb{R}$. Prove $a^2 + b^2 \geq 2ab$.

For each inequality, identify the nonnegative factor or square that makes the key step valid.

Practice 1.4: solution discussion

1. We know $x \geq 4 > 0$, so multiplying by x preserves $\geq$:

$$x^2 - 2x \geq 4x - 2x = 2x \geq 8.$$

2. Use the square of the difference:

$$a^2 + b^2 = (a - b)^2 + 2ab \geq 2ab.$$

The second result has no sign assumptions on a or b; the nonnegativity comes from $(a - b)^2$.

1.5 Abbreviate additive rearrangements

The book permits a short justification when a deduction needs only addition or subtraction of terms on the two sides.

```
example {x : Int} (h : x + 6 = 2) : x = -4 := by
  addarith [h]

example {u v : Real} (h : u <= 9 - v) : u + v < 10 := by
  addarith [h]
```

On paper: "From $u \leq 9 - v$, we get $u + v \leq 9 < 10$."

The course convention

For deductions requiring multiplication or division, keep the relevant calculation visible.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.5.

Practice 1.5: which steps are merely additive?

Assume the variables are real numbers.

1. $a - 2b = 3 \implies a = 2b + 3.$
2. $4w + 1 = 9 \implies w = 2.$
3. $m \leq 7 - n \implies m + n < 8.$

Explain each deduction. For the one that requires division, write a full chain.

Practice 1.5: solution discussion

1. Add $2b$ to both sides: an additive rearrangement.

$$a = 2b + 3.$$

2. Division by 4 is essential:

$$w = \frac{(4w + 1) - 1}{4} = \frac{9 - 1}{4} = 2.$$

3. Add n to both sides and compare constants:

$$m + n \leq 7 < 8.$$

The presence of $2b$ in the first statement does not mean that the deduction multiplies both sides.

Chapter 1: a checklist for calculation proofs

1. Identify the domain, hypotheses, and target endpoints.
2. Expose known expressions through algebraic identities.
3. Give a reason for each equality or inequality.
4. Check nonzero denominators and signs of multipliers.
5. Check that the relations combine in one direction.
6. Use Lean feedback to inspect individual steps.

Next chapter

A proof can branch, build intermediate facts, and introduce witnesses. Calculations will become components of larger arguments.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 1:

- 1.1 Proving equalities
- 1.2 Proving equalities in Lean
- 1.3 Tips and tricks
- 1.4 Proving inequalities
- 1.5 A shortcut

https://hrmacbeth.github.io/math2001/01_Proofs_by_Calculation.html

Companion code: https://github.com/hrmacbeth/math2001

These slides provide an independently written classroom presentation and new numerical practice. Use the book for additional examples and exercises.